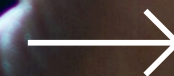


MLUP: Statistical Learning Theory

Presented by Gabriel Rodrigues Palma

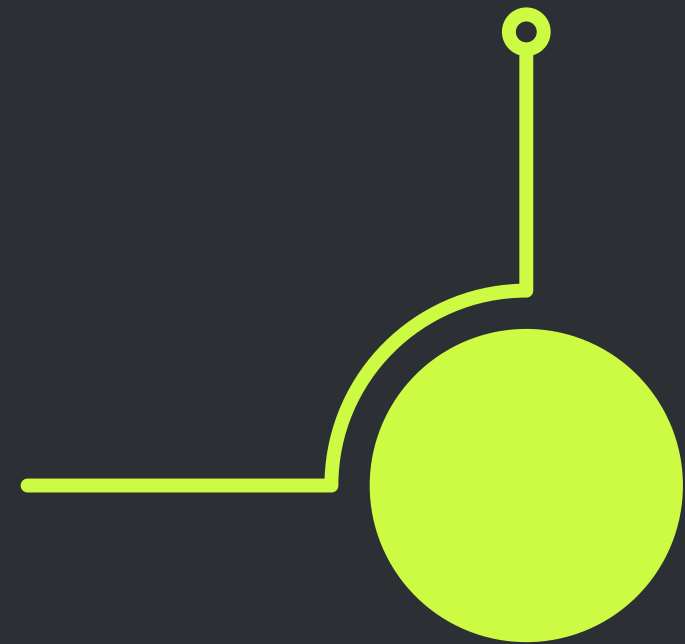


Machine Learning using
Python (MLUP01)



Day 5 (13:35 – 17:30)

SLT and Classification
(13:35 – 14:30)



Machine Learning using
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Vapnik's Contributions
(14:30 – 15:30)

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Vapnik's Contributions
(14:30 – 15:30)

SVM Overview
(16:30 – 17:30)

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Statistical Learning Theory: Models, Concepts, and Results

Ulrike von Luxburg
Max Planck Institute for Biological Cybernetics
Tübingen, Germany
`ulrike.luxburg@tuebingen.mpg.de`

Bernhard Schölkopf
Max Planck Institute for Biological Cybernetics
Tübingen, Germany
`bernhard.schoelkopf@tuebingen.mpg.de`

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1 Introduction

Statistical learning theory provides the theoretical basis for many of today's machine learning algorithms and is arguably one of the most beautifully developed branches of artificial intelligence in general. It originated in Russia in the 1960s and gained wide popularity in the 1990s following the development of the so-called *Support Vector Machine (SVM)*, which has become a standard tool for pattern recognition in a variety of domains ranging from computer vision to computational biology. Providing the basis of new learning algorithms, however, was not the only motivation for developing statistical learning theory. It was just as much a philosophical one, attempting to answer the question of what it is that allows us to draw valid conclusions from empirical data.

In this article we attempt to give a gentle, non-technical overview over the key ideas and insights of statistical learning theory. We do not assume that the reader has a deep background in mathematics, statistics, or computer science. Given the nature of the subject matter, however, some familiarity with mathematical concepts and notations and some intuitive understanding of basic probability is required. There exist many excellent references to more technical surveys of the mathematics of statistical learning theory: the monographs by one of the founders of statistical learning theory (Vapnik, 1995, Vapnik, 1998), a brief overview over statistical learning theory in Section 5 of Schölkopf and Smola (2002), more technical overview papers such as Bousquet et al. (2003), Mendelson (2003), Boucheron et al. (2005), Herbrich and Williamson (2002), and the monograph Devroye et al. (1996).

2 The standard framework of statistical learning theory

2.1 Background

In our context, learning refers to the process of inferring general rules by observing examples. Many living organisms show some ability to learn. For instance, children can learn what “a car” is, just by being shown examples of objects that are cars and objects that are not cars. They do not need to be told any rules about what is it that makes an object a car, they can simply learn the concept “car” by observing examples.

The field of machine learning does not study the process of learning in living organisms, but instead studies the process of learning in the abstract. The question is how a machine, a computer, can “learn” specific tasks by following specified learning algorithms. To this end, the machine is shown particular examples of a specific task. Its goal is then to infer a general rule which can both explain the examples it has seen already and which can generalize to previously unseen, new examples.

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What is Statistical Learning Theory?

- Which learning tasks can computers perform?
- What assumptions enable successful machine learning?
- What performance guarantees can we provide?

Core question: How can we draw valid conclusions from empirical data?

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SLT and Classification

The Learning Problem Setup

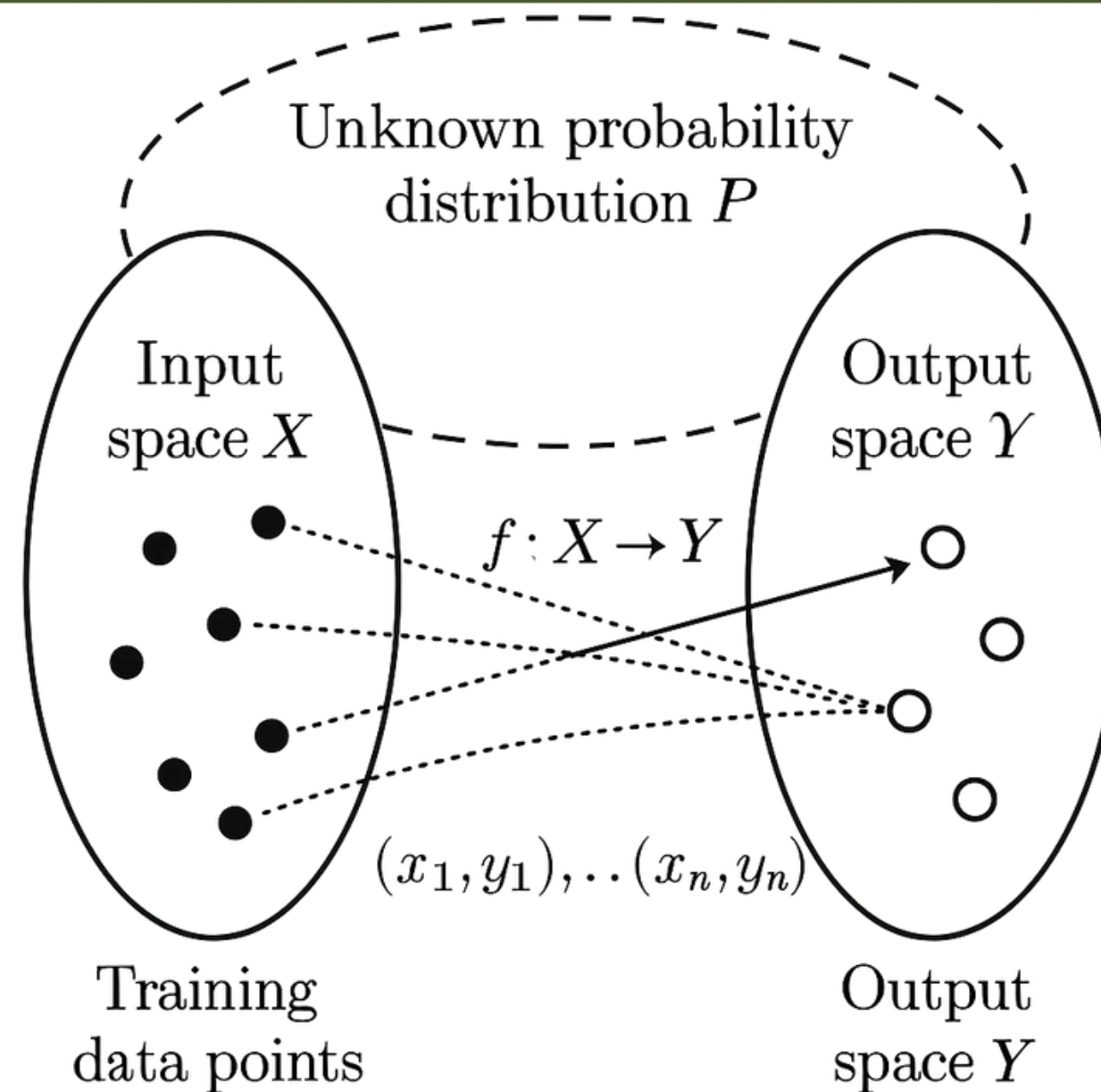
Input Space: $\mathcal{X}$ (space of instances/objects)

Output Space: $\mathcal{Y}$ (label space)

For binary classification: $\mathcal{Y} = \{-1, +1\}$

Goal: Find mapping $f : \mathcal{X} \rightarrow \mathcal{Y}$ that makes "as few errors as possible"

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Risk and Empirical Risk

True Risk (Expected Loss): $R(f) = E[\ell(X, Y, f(X))]$

Empirical Risk (Training Error): $R_{\text{emp}}(f) = \frac{1}{n} \sum_{i=1}^n \ell(x_i, y_i, f(x_i))$

Challenge: We want to minimize $R(f)$ but can only compute $R_{\text{emp}}(f)$

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The Bayes Classifier

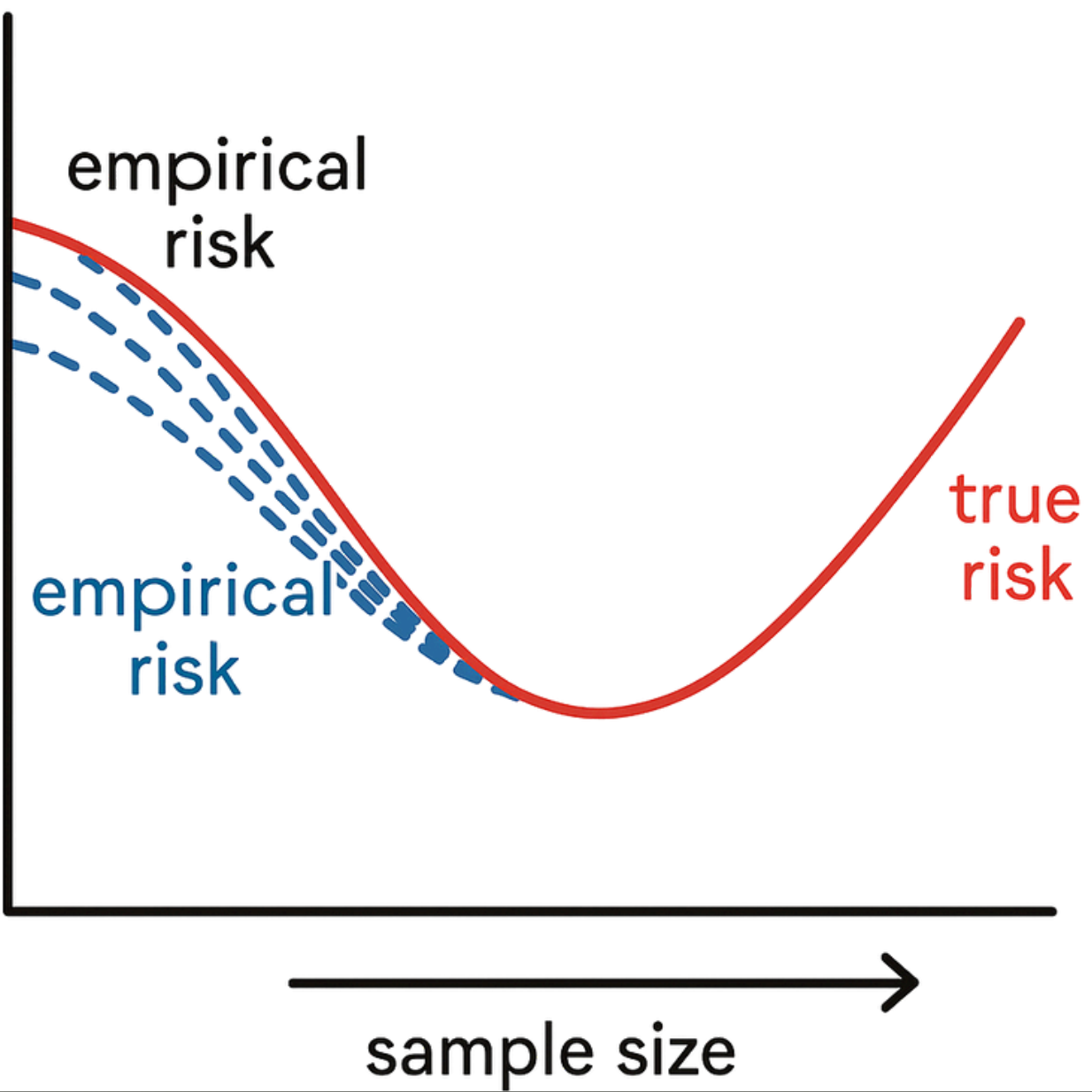
Conditional Probability: $\eta(x) := P(Y = 1 \mid X = x)$

Optimal Classifier:

$$f_{\text{Bayes}}(x) = \begin{cases} 1 & \text{if } P(Y = 1 \mid X = x) \geq 0.5 \\ -1 & \text{otherwise} \end{cases}$$

Problem: Cannot compute directly since P is unknown!

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Generalization and Consistency

Generalisation: A classifier f_n generalizes well if:

$|R(f_n) - R_{\text{emp}}(f_n)|$ is small

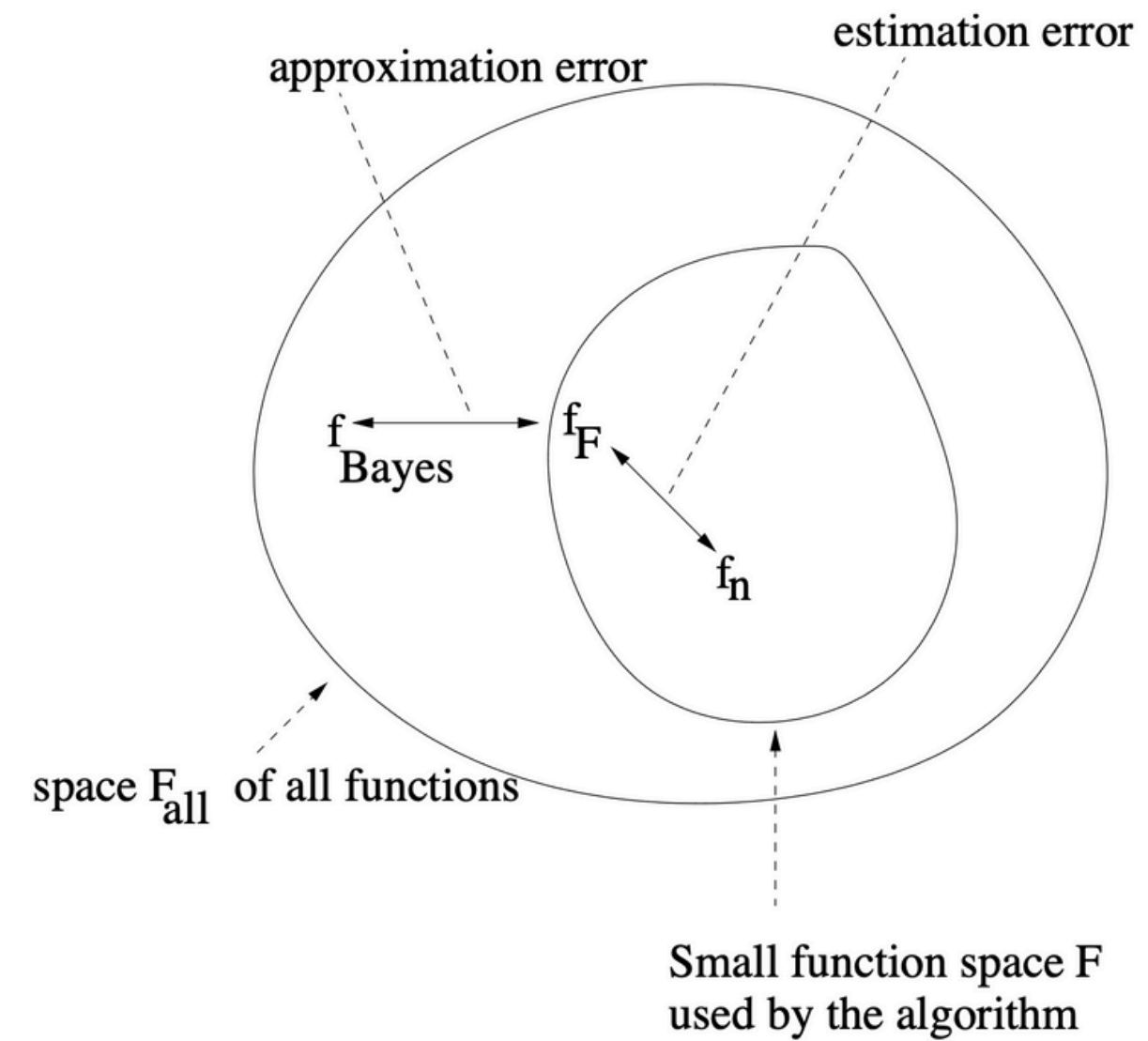
Consistency: Learning algorithm is consistent if:

$P(R(f_n) - R(f_{\mathcal{F}}) > \varepsilon) \rightarrow 0$ as $n \rightarrow \infty$

Bayes-Consistency:

$P(R(f_n) - R(f_{\text{Bayes}}) > \varepsilon) \rightarrow 0$ as $n \rightarrow \infty$

SLT and Classification



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